Consider a weighted least square problem:

$$min_b \frac{1}{N}\sum_{i=1}^{N} w_i(y_i - x_i'b)^2 \tag{1}$$

where w_i is the weight on unit i in the random sample. First order condition with respect to b:

$$\begin{aligned}\frac{\partial}{\partial b} &= -2\frac{1}{N}\sum_i w_i x_i(y_i - x_i'b) = 0 \\ \rightarrow \hat{\beta}_w &= \left(\frac{1}{N}\sum_i w_i x_i x_i'\right)^{-1}\left(\frac{1}{N}\sum_i w_i x_i y_i\right) \end{aligned}\tag{2}$$

Special case: suppose the units belong to either group a or group b. Consider $x_i = \begin{pmatrix} D_{ai} \\ D_{bi} \end{pmatrix}$ where $D_{ai} = 1[i \in a]$, and $D_{bi} = 1[i \in b]$. Now we have a weighted regressions with two dummy variables. You could get the FOC of the problem below:

$$min_b \frac{1}{N}\sum_{i=1}^{N} w_i(y_i - D_{ai}b_1 - D_{bi}b_2)^2 \tag{3}$$

FOC w.r.t. b_1:

$$\begin{aligned}\frac{\partial}{\partial b_1} &= -2\frac{1}{N}\sum_{i=1}^{N} w_i D_{ai}(y_i - D_{ai}b_1 - D_{bi}b_2) \\ &= -2\frac{1}{N}\left(\sum_{i\in a} w_i\, 1\,(y_i - b_1 - 0 * b_2) + \sum_{i\in b} 0\right) = 0 \\ \Rightarrow \hat{\beta}_1^w &= \frac{\sum_{i\in a} w_i y_i}{\sum_{i\in a} w_i}\end{aligned}\tag{4}$$

Note in the 2nd line of the FOC, we use the fact that $\sum_i w_i D_{ai} y_i = \sum_{i\in a} w_i y_i + \sum_{i\in b} w_i 0 y_i = \sum_{i\in a} w_i y_i$.
Similarly, $\hat{\beta}_2^w = \frac{\sum_{i\in b} w_i y_i}{\sum_{i\in b} w_i}$ represents the weighted average in group b.

Alternatively, you may want to verify (2) in vector form directly:

$$\begin{aligned}\frac{1}{N}\sum_i w_i x_i x_i &= \frac{1}{N}\sum_i w_i \begin{pmatrix} D_{ai} \\ D_{bi} \end{pmatrix}\begin{pmatrix} D_{ai} & D_{bi} \end{pmatrix} \\ &= \frac{1}{N}\sum_i w_i \begin{pmatrix} D_{ai}^2 & D_{ai}D_{bi} \\ D_{ai}D_{bi} & D_{bi}^2 \end{pmatrix} \\ &= \begin{pmatrix} \frac{1}{N}\sum_{i\in a} w_i & 0 \\ 0 & \frac{1}{N}\sum_{i\in b} w_i \end{pmatrix}\end{aligned}\tag{5}$$

using the fact that $D_{ai} * D_{bi} \equiv 0$. Since it is a diagonal matrix,

$$(\frac{1}{N}\sum_i w_i x_i x_i)^{-1} = \begin{pmatrix} (\frac{1}{N}\sum_{i\in a} w_i)^{-1} & 0 \\ 0 & (\frac{1}{N}\sum_{i\in b} w_i)^{-1} \end{pmatrix}$$

and the vector $\frac{1}{N}\sum_i w_i x_i y_i$:

$$\begin{aligned} \frac{1}{N}\sum_i w_i x_i y_i &= \frac{1}{N}\sum_i w_i \begin{pmatrix} D_{ai} \\ D_{bi} \end{pmatrix} y_i \\ &= \begin{pmatrix} \frac{1}{N}\sum_i w_i D_{ai} y_i \\ \frac{1}{N}\sum_i w_i D_{bi} y_i \end{pmatrix} \\ &= \begin{pmatrix} \frac{1}{N}\sum_{i\in a} w_i y_i \\ \frac{1}{N}\sum_{i\in b} w_i y_i \end{pmatrix} \end{aligned} \tag{6}$$

therefore,

$$\begin{aligned} \hat{\beta}^w =& (\frac{1}{N}\sum_i w_i x_i x_i)^{-1}(\frac{1}{N}\sum_i w_i x_i y_i) \\ =& \begin{pmatrix} (\frac{1}{N}\sum_{i\in a} w_i)^{-1} & 0 \\ 0 & (\frac{1}{N}\sum_{i\in b} w_i^{-1} \end{pmatrix} \begin{pmatrix} \frac{1}{N}\sum_{i\in a} w_i y_i \\ \frac{1}{N}\sum_{i\in b} w_i y_i \end{pmatrix} \\ =& \begin{pmatrix} \sum_{i\in a} w_i y_i / \sum_{i\in a} w_i \\ \sum_{i\in b} w_i y_i / \sum_{i\in b} w_i \end{pmatrix} \end{aligned} \tag{7}$$

which matches with our solution in (4). The first coefficient is the weighted average of y_i within group a, given weights w_i's. The second coefficient is the weighted average in group b.